

Artificial Intelligence II - Homework 3

Prompting-Based Response Clarity Classification

Andreas Georgiou
Academic ID: sdi2100218

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Abstract

In this report, a prompt-based classifier for response clarity classification on political question-answer pairs is proposed. The goal is to classify each response into one of three classes, namely Clear Reply, Ambivalent, and Clear Non-Reply, without fine-tuning the underlying language model. Rather, instruction-tuned Qwen/Qwen3.5 decoder-only models are utilized via deterministic prompting. The experimental setting involves model sizes (0.8B, 2B, and 4B) and prompting methods (zero-shot, few-shot, and chain-of-thought). The best configuration in terms of validation performance was Qwen/Qwen3.5-2B with zero-shot prompting and deterministic generation, yielding 59.71% accuracy, 0.401 macro-F1, and 0.545 weighted-F1 on a 690-instance validation dataset. It has been observed that prompt writing and model size have significant effects on the class bias. Specifically, few-shot prompting tends to collapse into the majority class, whereas chain-of-thought prompting is less stable.

1 Introduction

The purpose of this homework is to develop a response clarity classifier based on prompting technique rather than gradient-based fine-tuning. Each sample will have a political interview question and the respective answer. Classifier has to predict one of the following three classes:

- *Clear Reply*: the answer directly and specifically answers the question;
- *Ambivalent*: the answer is relevant but vague, partial, indirect, or incomplete;
- *Clear Non-Reply*: the answer avoids the question, changes topic, or does not answer it.

2 Dataset and Validation Setup

The loaded dataset has a total of 3,448 examples in the training dataset and 308 examples in the testing dataset. Stratified splitting was performed on the labeled dataset to create the pool/validation split, with 690 examples in the validation set, leaving 2,758 examples in the pool.

Table 1: Class distribution in the full labeled training set.

Class	Count	Percentage
<i>Ambivalent</i>	2040	59.16%
<i>Clear Reply</i>	1052	30.51%
<i>Clear Non-Reply</i>	356	10.32%

Table 2: Class distribution in the validation set.

Class	Count	Percentage
<i>Ambivalent</i>	408	59.13%
<i>Clear Reply</i>	211	30.58%
<i>Clear Non-Reply</i>	71	10.29%

The class distribution is evidently unbalanced, with “Ambivalent” being the largest class and “Clear Non-Reply” being the smallest class. Accuracy would therefore be inadequate for assessing the results because of this class imbalance. Macro-F1 is critical in this case since it treats all classes equally, and weighted-F1 is included for the same reason. The best system was selected based on weighted-F1 score in the final run, but macro-F1 score is always presented.

3 Prompting Methodology

3.1 Models

The main experiments used the Qwen3.5 instruction-tuned model family:

- Qwen/Qwen3.5-0.8B,
- Qwen/Qwen3.5-2B,
- Qwen/Qwen3.5-4B.

These models were kept frozen. No gradient-based fine-tuning was performed. All classification decisions were produced through prompting and output parsing.

3.2 Prompt Format

The final prompt uses a strict but balanced definition of the three classes. The most important design choice was to separate *Ambivalent* from *Clear Non-Reply*. Earlier prompts were too strict and caused the model to overpredict *Clear Non-Reply*. The final system prompt therefore used the rule: if the answer is related but incomplete, vague, indirect, or only partially answers the question, choose *Ambivalent*; choose *Clear Non-Reply* only when the answer mostly or completely avoids the question.

The final zero-shot user prompt had the following structure:

```
Classify the following political question-answer pair.

Question:
<question>

Answer:
<answer>

Valid labels:
Clear Reply
Ambivalent
Clear Non-Reply

Decision guide:
- Choose Clear Reply only if the answer directly answers the exact question.
- Choose Ambivalent if the answer is relevant but partial, vague, indirect, incomplete
  , or mixed.
- Choose Clear Non-Reply only if the answer mostly or completely avoids the question.
- If unsure between Ambivalent and Clear Non-Reply, choose Ambivalent.
```

Final label:

3.3 Prompting Strategies

Three prompting strategies were evaluated:

Zero-shot The model receives the instruction, definitions, question, answer, and output label indicator, but no labeled demonstrations.

Few-shot The prompt includes balanced labeled demonstrations from all three classes. Two examples per class were selected from the prompt pool using a fixed random seed.

Chain-of-thought The prompt asks the model to briefly consider whether the answer directly answers, partially answers, or avoids the question, but the output is still constrained to a final label format.

3.4 Inference and Parsing

Generation was deterministic, with sampling disabled. The model was instructed to return only the final label. The parser searched for the three valid label strings and normalized minor variations such as hyphenation in *Clear Non-Reply*. A constrained label-scoring approach was also explored during development, but it produced strong class bias, especially against *Ambivalent*. Therefore, it was not used as the final inference mode. The final experiments use deterministic generation with strict parsing.

4 Experimental Results

4.1 Overall Comparison

Table 3 shows the full validation results. The best configuration was Qwen/Qwen3.5-2B with zero-shot prompting.

Table 3: Validation results for all model and prompting-strategy combinations. Results are sorted by weighted-F1.

Model	Strategy	Accuracy	Macro-P	Macro-R	Macro-F1	Weighted-F1
Qwen3.5-2B	zero-shot	0.597	0.479	0.403	0.401	0.545
Qwen3.5-0.8B	zero-shot	0.590	0.530	0.333	0.250	0.441
Qwen3.5-0.8B	few-shot	0.591	0.197	0.333	0.248	0.439
Qwen3.5-2B	few-shot	0.591	0.197	0.333	0.248	0.439
Qwen3.5-4B	few-shot	0.591	0.197	0.333	0.248	0.439
Qwen3.5-2B	CoT	0.439	0.411	0.439	0.385	0.422
Qwen3.5-0.8B	CoT	0.406	0.404	0.368	0.314	0.375
Qwen3.5-4B	zero-shot	0.412	0.201	0.263	0.225	0.369
Qwen3.5-4B	CoT	0.148	0.297	0.351	0.113	0.106

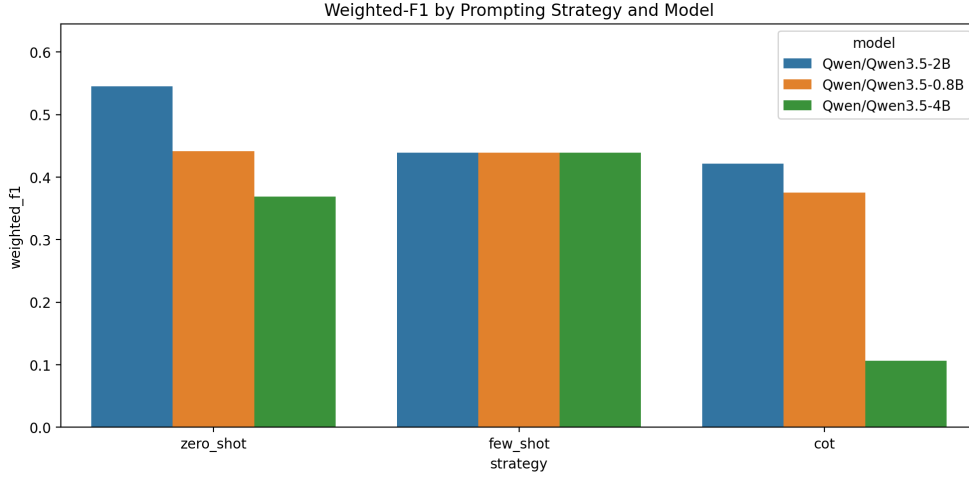


Figure 1: Weighted-F1 by prompting strategy and model size. The best result is obtained by Qwen3.5-2B with zero-shot prompting.

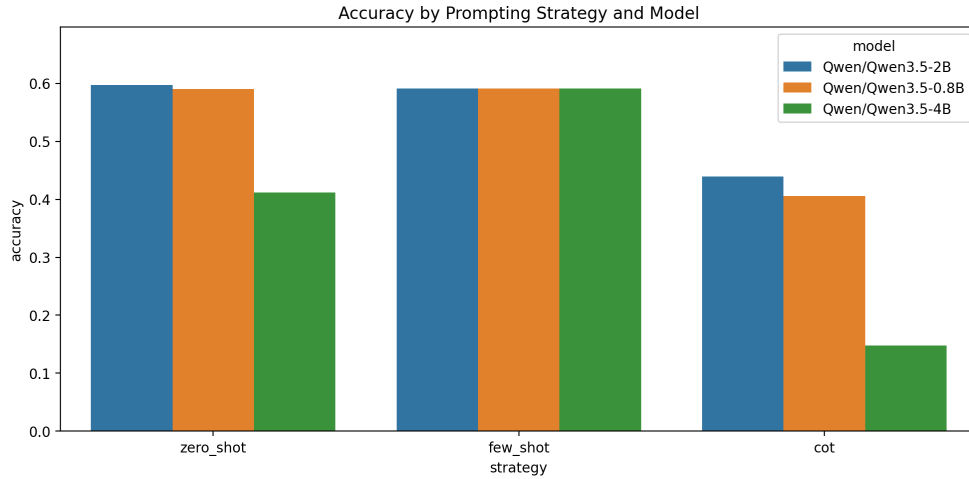


Figure 2: Accuracy by prompting strategy and model size. Accuracy alone is misleading for few-shot prompting because the model often predicts the majority class.

4.2 Best System

The best system was Qwen/Qwen3.5-2B with zero-shot prompting and deterministic generation. Its validation results are:

- accuracy: 0.597,
- macro-F1: 0.401,
- weighted-F1: 0.545,
- weighted precision: 0.562,
- weighted recall: 0.597.

Table 4 displays the class-based analysis of our experiment results. The model works best with the Ambivalent category, which is the most frequent class in our dataset. The model demonstrates moderate precision but low recall for Clear Reply, which means that many clear replies are categorized as Ambivalent. The least effective category is

Table 4: Class-wise performance of the best validation system.

Class	Precision	Recall	F1	Support
<i>Clear Reply</i>	0.55	0.24	0.34	211
<i>Ambivalent</i>	0.62	0.87	0.72	408
<i>Clear Non-Reply</i>	0.27	0.10	0.14	71
Macro average	0.48	0.40	0.40	690
Weighted average	0.56	0.60	0.55	690

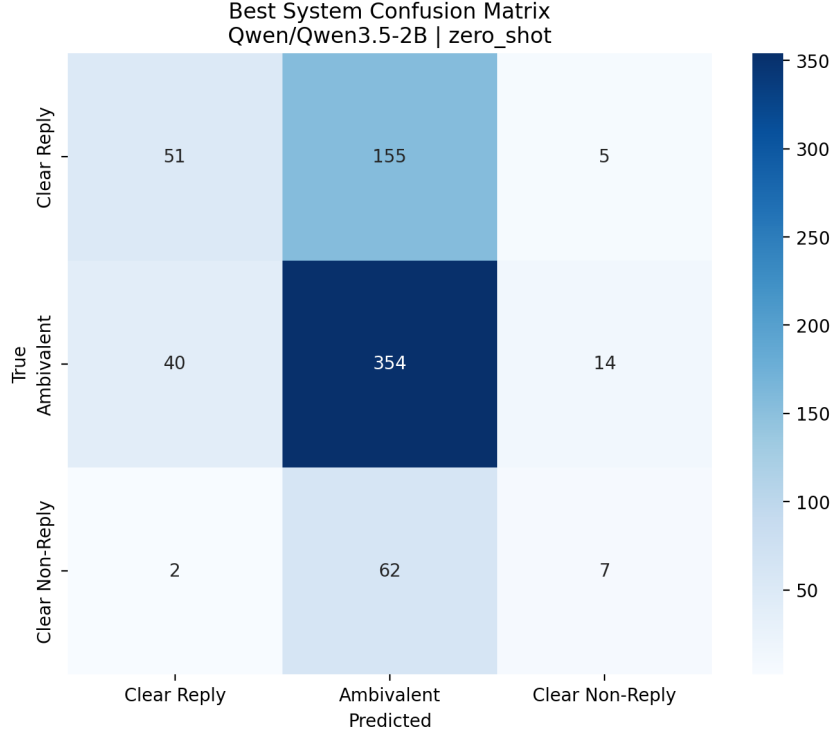


Figure 3: Confusion matrix for the best system. The main error pattern is that both direct replies and clear non-replies are often mapped to *Ambivalent*.

The confusion matrix shows the following counts:

- 354 of 408 *Ambivalent* examples were correctly classified;
- 51 of 211 *Clear Reply* examples were correctly classified, while 155 were predicted as *Ambivalent*;
- 7 of 71 *Clear Non-Reply* examples were correctly classified, while 62 were predicted as *Ambivalent*.

This confirms that the final prompt is conservative: it usually prefers *Ambivalent* for borderline cases. This improves weighted-F1 and avoids the earlier overprediction of *Clear Non-Reply*, but it reduces recall for *Clear Reply* and *Clear Non-Reply*.

4.3 Effect of Prompting Strategy

From the above experiments, it can be concluded that the zero-shot prompting approach worked out better. The Qwen3.5-2B zero-shot model performed the best in terms of both weighted-F1 and macro-F1 in comparison with other models. Few-shot prompting showed comparable

performance for all sizes of models; however, the value of macro-F1 was only 0.248. It means majority class collapse because the model predicted the Ambivalent category frequently, and since Ambivalent is the largest class, it provided good results.

In addition, chain-of-thought prompting was not useful for creating the final system. Qwen3.5-2B CoT had a macro-F1 equal to 0.385, which is close to the highest zero-shot macro-F1; however, the accuracy and weighted-F1 of Qwen3.5-2B CoT were worse. Moreover, Qwen3.5-4B CoT was particularly unstable. The possible reason is that a chain-of-thought prompt makes the model think longer, but it does not help to produce better predictions. In this case, clear instructions about labels work better than chain-of-thought prompting.

4.4 Effect of Model Size

An increase in model size did not result in monotonous improvement. The 2B model was the most accurate, while the 4B model was underperforming in both zero-shot and CoT conditions. This indicates that prompt sensitivity and output behavior were more important factors than model size in this case. It might be that the 4B model followed the strict prompt and hence made overly cautious or unstable predictions. The 0.8B model had acceptable accuracy but low macro-F1, indicating that the smaller model struggled with balancing all three labels.

4.5 Effect of validation set size

Moreover, it can be seen from the final Kaggle results that having a larger number of validation examples indeed aided model selection. The smallest debug run, which was conducted with only 90 stratified validation examples, produced the lowest score among all attempts, which was equal to 0.46321. The increase in the number of examples in the debug subset up to 300 led to the improvement of the result to 0.53999. The best result of 0.54461 was produced by the experiment with the full validation set. While the use of 90 examples ensured the maintenance of the global class distribution, it was still insufficient to reflect the diversity of question-answer interactions. Thus, the final system was selected with the help of the full validation set.

Setting	Validation size	Kaggle score
Debug small	90	0.46321
Debug medium	300	0.53999
Full validation	690	0.54461

Table 5: Effect of validation size on the submitted system.

5 Subgroup Analysis

The validation set was divided into subgroups according to question length and answer length. Table 6 reports the performance of the best system.

Table 6: Subgroup analysis by question and answer length for the best system.

Group type	Group	N	Accuracy	Macro-F1	Weighted-F1
Question length	short questions	237	0.536	0.393	0.483
Question length	medium questions	224	0.612	0.334	0.525
Question length	long questions	229	0.646	0.459	0.624
Answer length	short answers	230	0.504	0.385	0.447
Answer length	medium answers	230	0.587	0.299	0.492
Answer length	long answers	230	0.700	0.428	0.684

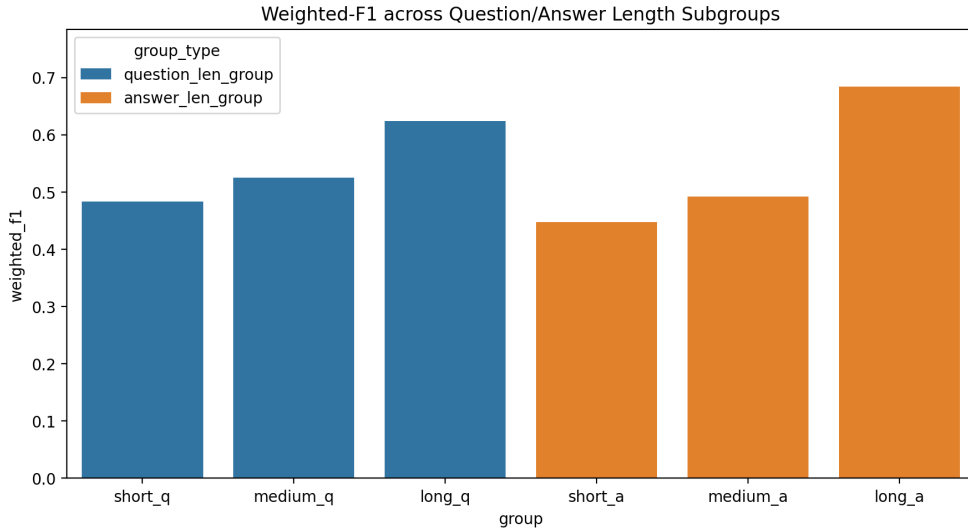


Figure 4: Weighted-F1 across question-length and answer-length subgroups. Long-answer examples obtain the highest weighted-F1, while short-answer examples are harder.

The best performance appears for long questions and long answers. Long answers may provide more evidence about whether the answer is direct, partial, or evasive. Short answers are harder because there is less context; a very short answer can be a direct reply, a refusal, or an evasive fragment depending on the question. The model also performs better on long questions, possibly because detailed questions make the expected answer clearer.

6 Comparison with Previous Assessments

Table 7 compares the prompting-based system with previous vector-based and encoder-only approaches. The previous numbers are included as reference values from earlier assessments.

Table 7: Comparison with previous assessment systems.

Model	Accuracy	Weighted-F1
TF-IDF + Logistic Regression	0.650	0.570
BERT encoder-only fine-tuning	0.670	0.673
Qwen3.5-2B zero-shot prompting	0.597	0.545

The prompting model underperforms the previously described supervised encoder-only approach. That is understandable since the BERT model training includes gradient update based

on the labeled training data, whereas the prompting model employs the Qwen model without any update in the frozen inference-only mode. Similarly, the TF-IDF classifier remains quite efficient because the classification task may rely on clear lexical and structural patterns in the response texts. However, the strength of the prompting model is its independence from training: it does not require any training process, which can be adjusted just by changing the prompt. The failure modes vary. While the prompting model excessively relies on *Ambivalent* in the final version, previous supervised models learned more complex dataset-specific boundaries. That indicates that the prompting approach is capable of identifying the general semantics relevance but fails to define the specific boundary between partial answers and non-answers.

7 Error Analysis

The main problem pattern in this case is overprediction of *Ambivalent*. For example, in the best-performing system, 155 *Clear Reply* samples and 62 *Clear Non-Reply* samples were incorrectly predicted as *Ambivalent*. This means that the prompt succeeded in avoiding overuse of *Clear Reply*, but it became too conservative. There are three recurring types of problems found:

1. **Direct but lengthy answers that were classified as ambiguous.** Political answers can be lengthy, including additional information about the subject, background, or qualifications. If the answer ultimately responds to the question, the model may still consider its indirect style and classify it as *Ambivalent*.
2. **Very short answers which seem evasive.** Some very short answers seem evasive but were classified as *Ambivalent* in the dataset. However, the model classified them as *Clear Non-Reply* since they appear evasive in isolation.
3. **Answers that are topically relevant but evasive.** Some answers contain topics relevant to the question but do not address it directly. This problem was addressed by the strict prompt but not completely resolved.

7.1 Concrete Examples

These patterns are illustrated by the error examples. In one instance, the question concerned the policy towards Kashmir and whether the US can be involved. The gold label was *Clear Reply*, but the model predicted *Ambivalent*. The reply concerned Kashmir and the possibility for US involvement if the sides agreed to do so, but as it was diplomatic and qualified, the model regarded it as incomplete. In another example, the question was whether the schools needed to be ready for the spread of coronavirus. The gold label was *Clear Reply*, but the model predicted *Ambivalent*. The reply started with a straightforward affirmative statement but then continued to hedge and provide context. This indicates that the model punishes hedging too much. *Ambivalent* errors include an example with a question asking about posing another question, which was answered with a short explanation. The prediction was *Clear Non-Reply*. It shows how hard it is to analyze conversational fragments without additional discourse context.

8 Discussion

From the experiments, we can see that the prompt design plays the most important role in this setup. At first, the strict prompts tended to overpredict *Clear Non-Reply*, whereas soft prompts overpredicted *Ambivalent*. In our final prompt, we try to achieve an equilibrium between the two approaches by using a definition where *Ambivalent* refers to related but non-complete answers, while *Clear Non-Reply* means mostly evasive answers. We can also see from our experiments that large models are not necessarily better in this scenario. Our 2B model performed better than the other two (0.8B and 4B). This could be caused by the interaction

between the instruction-following property of the model and the prompt itself. Perhaps, the 4B model needs some special generation parameters or a different prompt. A possible future improvement would be to use a combination of prompting and calibration/self-consistency. The model could be asked whether the answer is: (1) relevant, (2) directly answering the question, (3) mostly evasive? Another useful improvement would be retrieval of semantically similar examples.

9 Conclusion

In this study, a prompting-based classifier for response clarity classification was designed and tested. The best-performing system employed Qwen/Qwen3.5-2B with zero-shot prompting and deterministic generation. It yielded an accuracy of 0.597, a macro-F1 score of 0.401, and a weighted-F1 score of 0.545 on the validation set. Zero-shot prompting outperformed few-shot and chain-of-thought prompting in the final experiments. Class-boundary calibration was the biggest drawback, as the model often mistakenly identified clear replies and clear non-replies as the ambiguous class. In general, the experiments show that prompting can yield a useful classifier, yet encoder-only models trained through supervision still outperform them.